

Aiman Hazim

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Education

- University of Fukui**, B.Eng. Mechanical and System Engineering (Robotics Course) Expected Mar 2027
- Fully sponsored by the Malaysian government scholarship program for studies in Japan
 - Relevant coursework: Autonomous Systems, Robot Control, Machine Inference, Graphics and Recognition, Robot Mechanism, Signal Processing, Robots and Dynamical Systems, Applied Mathematics
- University of Malaya**, Preparatory Program for Study in Japan (Special Course) Feb 2021

Research Experience

- Undergraduate Researcher**, Intelligent Robot Laboratory, University of Fukui Oct 2025 – Present
- Thesis: Large-Scale Simulation with Human-Behavior AI for TIR-LiDAR Robot Navigation*
- Deploying Isaac Sim 5.1.0 on a remote cloud GPU instance, independently resolving Docker permission, SSH configuration, and GPU streaming issues to enable an AI-assisted development workflow
 - Building a ROS data pipeline on the Mega Rover mobile robot: synchronized multi-sensor recording with rosbag, human detection from TIR imagery via thermal thresholding, and depth-image generation from Velodyne VLP-16 point clouds
 - Designing a 20 m × 20 m dynamic indoor environment in NVIDIA Isaac Sim populated with multiple AI-driven human agents, each with individual appearance, gait characteristics, and collision-avoidance behavior, to evaluate SLAM performance statistically across 100+ environment configurations
 - Implementing active multimodal SLAM fusing thermal-infrared (TIR) camera and 3D LiDAR data to achieve robust localization and traversability mapping in dark and crowded environments, extending prior lab work (Walk2Map++/HO3-SLAM) from passive to active vision

Projects

- PyroScout, Thermal-LiDAR Fusion for Search-and-Rescue** (personal project) Jun 2026
- Built a from-scratch 2D autonomous search-and-rescue simulator in Python that fuses 2D LiDAR for geometry and a thermal sensor for semantics, letting a robot map an unknown building and locate a heat-emitting victim with no prior map
 - Implemented the full classical autonomy stack: log-odds occupancy-grid mapping, frontier-based exploration, clearance-aware A* planning, and a pure-pursuit controller, coordinated by a behavior state machine
 - Achieved 8/8 successful rescues with zero wall collisions across randomized sensor-noise trials, localizing the victim to ~1 cm via thermal-LiDAR fusion, covered by 41 unit/integration tests with CI
- Self-Hosted LLM & RAG Assistant** (personal project) May 2025
- Built a fully offline, private RAG assistant with llama.cpp (GGUF) and Open WebUI, indexing personal notes and research papers for on-device retrieval and tuning quantization and GPU offload to run capable models within local memory limits
- Autonomous Vacuum Robot**, Robot Engineering Creative Experiment course (team of 3) Jan 2024
- Designed and machined the acrylic robot body in Onshape, built the electronics from KiCad circuit design through soldering and assembly, and programmed control logic on an Arduino UNO R3
 - Received the Eiwa System Management Award at the final assessment, where each attending company selected its top team, for improvising a design fix on the spot to clear the challenge course

Work Experience

- Newspaper Delivery Staff (part-time)**, Fukui Shimbun – Fukui, Japan Sep 2025 – Present
- Completed early-morning delivery routes alongside thesis research
- Inventory Auditor (part-time)**, AJIS Co., Ltd. – Fukui, Japan Jun 2025 – Present
- Performed high-accuracy stocktaking counts for retail clients as part of audit teams under fixed time windows
- Housekeeping Staff & New-Staff Trainer (part-time)**, Hotel Fujita – Fukui, Japan Nov 2022 – Present
- Trained and onboarded new staff members over 3.5+ years of service alongside full-time engineering studies
 - Maintained room-quality standards under strict per-room time constraints by refining and standardizing cleaning procedures

Skills

Programming: Python, C/C++

Robotics & Tools: ROS, NVIDIA Isaac Sim, Velodyne 3D LiDAR, thermal (TIR) cameras, Docker, Linux, Git

Hardware & CAD: Arduino, KiCad, circuit design and soldering, Onshape

Languages: Malay (Native) • English (Fluent) • Japanese (Business)

Certifications

- TOEIC L&R**, 970/990 Jun 2026
- Driver's License (Japan, Class 1)** Jul 2022
- JLPT N2**, 173/180 Jan 2021